Stratified Sampling

SurvMeth/Surv 625: Applied Sampling

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Stratified sampling

- Implementation
- Inference
- Projection

Implementation

- Dividing our population of elements into subgroups (strata) using auxiliary information that is available *prior* to drawing the sample
- Simple random sampling of elements WITHIN each of the strata (or population subgroups): **Independent** across strata
- Need the auxiliary information on the frame to create mutually exclusive and exhaustive subgroups (strata)

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1. Avoid selecting a really bad SRS sample
2. Desire precision for subgroups
3. More convenient to administer and may results in a lower survey cost
4. Often gives more precise estimates for population means and totals

Inference

- We can apply everything that we've learned about for SRS within each of the strata
- Stratum index: $h = 1, \dots, H$
- Denote the variable of interest for i-th element in stratum h as Y_{hi}
- For each population stratum, population mean $\bar{Y}_h = \sum_{i=1}^{N_h} Y_{hi}/N_h$ and element variance $S_h^2 = \frac{1}{N_h - 1}\sum_{i=1}^{N_h}(Y_{hi} - \bar{Y}_h)^2$
- For each stratum in the sample, we can compute $\bar{y}_h, s_h^2, t_h, N_h, n_h$, etc., in addition to sampling variances, etc.; **all specific to h!**

Inference: Population mean

- We can rewrite the population mean as a weighted sum of the population means for each stratum, where the weight W_h is the relative proportion of the population within each stratum

$$\bar{Y} = \sum_h \frac{N_h}{N} \bar{Y}_h \doteq \sum_h W_h \bar{Y}_h$$

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- We can write the sample mean in the same way, assuming that we have good (unbiased) estimates of the means in each stratum

$$\bar{y}_w = \sum_h W_h \bar{y}_h$$

Inference: Sampling weight

- Element-level weighting:

$$\bar{y}_w = \frac{\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} y_{hi}}{\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi}}$$

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- For stratified sampling, $\pi_{hi} = n_h/N_h$, so we have $w_{hi} = N_h/n_h$ and $\sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} = N$.

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- A stratified sample is self-weighting is the sampling fraction n_h/N_h is the same across strata, where the sampling weight for each observation is N/n, exactly the same as in SRS.

Inference: Population mean cont.

Within stratum h: SRS

- Sampling fraction: $f_h = n_h/N_h$

Inference: Population mean cont.

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Within stratum h: SRS

- Sampling fraction: $f_h = n_h/N_h$
- Mean estimate: $\bar{y}_h$ [or p_h if proportion]
- Element variance estimate: $s_h^2 = \frac{1}{n_h-1}\sum_{i=1}^{n_h}(y_{hi} - \bar{y}_h)^2$ [or $s_h^2 = \frac{n_h}{n_h-1}p_h(1-p_h)$ if proportion]

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- Sampling variance estimate: $var(\bar{y}_h) = (1 - f_h) \frac{s_h^2}{n_h}$

Inference: Population mean cont.

Within stratum h: SRS

- Sampling fraction: $f_h = n_h/N_h$
- Mean estimate: $\bar{y}_h$ [or p_h if proportion]
- Element variance estimate: $s_h^2 = \frac{1}{n_h-1}\sum_{i=1}^{n_h}(y_{hi}-\bar{y}_h)^2$ [or $s_h^2 = \frac{n_h}{n_h-1}p_h(1-p_h)$ if proportion]
- Sampling variance estimate: $var(\bar{y}_h) = (1-f_h)\frac{s_h^2}{n_h}$
- Standard error: $se(\bar{y}_h) = \sqrt{var(\bar{y}_h)}$

Inference: Population mean cont.

Combine across strata

The sampling variance of the overall estimated mean is entirely a function of the within-stratum sampling variances only!

$$var(\bar{y}_w) = var(\sum_h W_h \bar{Y}_h) = \sum_h var(W_h \bar{Y}_h)$$

$$= \sum_h W_h^2 var(\bar{y}_h) = \sum_h W_h^2 (1 - f_h) \frac{s_h^2}{n_h} \quad (1)$$

Example: Estimating the average number of farm acres per county

Use four U.S. census regions as strata to select counties

Region	# Counties in population	# Counties in sample	Sample mean in region	Sample variance in region
Northest	220	21	?	?
North Central	1054	103	?	?
South	1382	135	?	?
West	422	41	?	?
Total	3178	300		

R code: Lu-Lohr-ch03.R

```
########## Selecting a Stratified Random Sample ##########

##### Example 3.2, Select a stratified sample from agpop

### Use the sample function with each stratum
popsize <- table(agpop$region)
# Select an SRS without replacement from each region with pre-set allocation
# with total size n=300
regionname <- c("NC","NE","S","W")
# Make sure sampsize has same ordering as regionname
sampsize <- c(103,21,135,41)
# Set the seed for random number generation
set.seed(108742)
index <- NULL
for (i in 1:length(sampsize)) {
  index <- c(index,sample((1:N)[agpop$region==regionname[i]],
                          size=sampsize[i],replace=F))
}
strsample<-agpop[index,]
# Check that we have the correct stratum sample sizes
table(strsample$region)
```

```
 NC  NE   S   W
103  21 135  41
```

```
# Print the first six rows of the sample to see
#strsample[1:6,]
```

R code cont.

```
### Use the strata function from the sampling package

# Sort the population by stratum
agpop2<-agpop[order(agpop$region),]
# Use the strata function to select the units for the sample
# Make sure size argument has same ordering as the stratification variable
index2<-strata(agpop2,stratanames=c("region"),size=c(103,21,135,41),
               method="srswor")
table(index2$region)  # look at number of counties selected within each region
```

```
 NC  NE   S   W
103  21 135  41
```

```
#head(index2)
strsample2<-getdata(agpop2,index2) # extract the sample
#head(strsample2)
```

R code cont.

```
# Calculate the sampling weights
# First check that no probabilities are 0
sum(strsample2$Prob<=0)
```

```
[1] 0
```

```
strsample2$sampwt<-1/strsample2$Prob
# Check that the sampling weights sum to the population sizes for each stratum
tapply(strsample2$sampwt,strsample2$region,sum)
```

```
  NC   NE    S    W
1054  220 1382  422
```

```
########## Computing Statistics from a Stratified Random Sample ##########

##### Example 3.2 and 3.6

data(agstrat)
#names(agstrat)  # list the variable names
#agstrat[1:6,1:8] # take a look at the first 6 obsns from columns 1 to 8
#nrow(agstrat)  # number of rows, 300
#unique(agstrat$region) # take a look at the four regions, NC, NE, S, W
table(agstrat$region)  # number of counties in each stratum
```

```
 NC  NE   S   W
103  21 135  41
```

```
# check that the sum of the weights equals the population size
sum(agstrat$strwt)  #3078
```

```
[1] 3078
```

R code cont.

```
### Draw a boxplot of the stratified random sample

#boxplot(acres92/10^6 ~ region, xlab = "Region", ylab = "Millions of Acres",
#        data = agstrat)
# notice the large variability in western region
```

R code: *survey* package

```
### Set up information for the survey design

# create a variable containing population stratum sizes, for use in fpc (optional)
# popsize_recode gives popsize for each stratum
popsize_recode <- c('NC' = 1054, 'NE' = 220, 'S' = 1382, 'W' = 422)
# next statement substitutes 1054 for each 'NC', 220 for 'NE', etc.
agstrat$popsize <- popsize_recode[agstrat$region]
table(agstrat$popsize)  #check the new variable
```

```
 220  422 1054 1382
  21   41  103  135
```

```
# input design information for agstrat
dstr <- svydesign(id = ~1, strata = ~region, weights = ~strwt, fpc = ~popsize,
                  data = agstrat)
#dstr

### Calculate the statistics using the design object

# calculate mean, SE and confidence interval
smean<-svymean(~acres92, dstr);smean
```

```
          mean    SE
acres92 295561 16380
```

```
confint(smean, level=.95, df=degf(dstr)) # note that df = n-H = 300-4
```

```
         2.5 %   97.5 %
acres92 263325 327796.5
```

```
# calculate total, SE and CI
stotal<-svytotal(~acres92, dstr); stotal
```

R code: *survey* package cont.

```
### Alternative design specifications

# Get same result if omit weights argument since weight = popsize/n_h
dstrfpc <- svydesign(id = ~1, strata = ~region, fpc = ~popsize, data = agstrat)
svymean(~acres92, dstrfpc)
```

```
          mean    SE
acres92 295561 16380
```

```
# If you include weights but not fpc, get SE without fpc factor
dstrwt <- svydesign(id = ~1, strata = ~region, weights = ~strwt, data = agstrat)
svymean(~acres92, dstrwt)
```

```
          mean    SE
acres92 295561 17241
```

```
### Calculating stratum means and variances

# calculate mean and se of acres92 by regions
svyby(~acres92, by=~region, dstr, svymean, keep.var = TRUE)
```

```
   region   acres92       se
NC     NC 300504.16 16107.59
NE     NE  97629.81 18149.49
S       S 211315.04 18925.35
W       W 662295.51 93403.65
```

```
# calculate total and se of acres92 by regions
svyby(~acres92, ~region, dstr, svytotal, keep.var = TRUE)
```

```
   region   acres92       se
NC     NC 316731380 16977399
NE     NE  21478558  3992889
S       S 292037391 26154840
```

R code: *survey* package cont.

```
### Estimating proportions from a stratified random sample

# Option 1: Use a 0-1 variable and find its mean
# Create variable lt200k
agstrat$lt200k <- rep(0,nrow(agstrat))
agstrat$lt200k[agstrat$acres92 < 200000] <- 1
# Rerun svydesign because the data set now has a new variable
dstr <- svydesign(id = ~1, strata = ~region, fpc = ~popsize,
  weights = ~strwt, data = agstrat)
# calculate proportion, SE and confidence interval
smeanp<-svymean(~lt200k, dstr); smeanp
```

```
          mean     SE
lt200k 0.51391 0.0248
```

```
confint(smeanp, level=.95,df=degf(dstr))
```

```
           2.5 %    97.5 %
lt200k 0.4651188 0.5627107
```

```
# calculate total, SE and CI
stotalp<-svytotal(~lt200k, dstr); stotalp
```

```
        total     SE
lt200k 1581.8 76.318
```

```
confint(stotalp, level=.95,df=degf(dstr))
```

```
          2.5 %   97.5 %
lt200k 1431.636 1732.024
```

```
# Option 3: Construct asymmetric confidence intervals
# calculate proportion and confidence interval with svyciprop
svyciprop(~I(lt200k==1), dstr, method="beta")
```

Analysis of variance (ANOVA)

The sum of squares

$$
\begin{aligned}
\sum_{h=1}^{H}\sum_{i=1}^{N_h}(Y_{hi}-\bar{Y})^2 &= (N-1)S^2 \\
&= \sum_{h=1}^{H}\sum_{i=1}^{N_h}(Y_{hi}-\bar{Y}_h+\bar{Y}_h-\bar{Y})^2 \\
&= \sum_{h=1}^{H}\sum_{i=1}^{N_h}(Y_{hi}-\bar{Y}_h)^2+\sum_{h=1}^{H}\sum_{i=1}^{N_h}(\bar{Y}_h-\bar{Y})^2 \\
&= \sum_{h=1}^{H}(N_h-1)S_h^2+\sum_{h=1}^{H}N_h(\bar{Y}_h-\bar{Y})^2 \\
SSTO &\doteq SSW+SSB
\end{aligned}
$$

ANOVA cont.

We have the simplification as

$$
\begin{aligned}
S^2 &= \sum_{h=1}^{H} \frac{N_h - 1}{N - 1} S_h^2 + \sum_{h=1}^{H} \frac{N_h - 1}{N - 1} (\bar{Y}_h - \bar{Y})^2 \\
&\approx \sum_{h=1}^{H} W_h S_h^2 + \sum_{h=1}^{H} W_h (\bar{Y}_h - \bar{Y})^2 \qquad (2) \\
&= \text{Within-stratum variance} + \text{Between-stratum variance}
\end{aligned}
$$

- The overall S^2 is fixed; if we define strata such that the between-stratum variance component becomes large, the within-stratum variance will necessarily become smaller

ANOVA cont.

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- The overall S^2 is fixed; if we define strata such that the between-stratum variance component becomes large, the within-stratum variance will necessarily become smaller
- Hence the sampling variance will go down based on Equation (1), which only depends on the within-stratum variance
- Decrease the sampling variance of the mean by **making strata heterogeneous between and homogeneous within**

Projection

- Always stratify! We give ourselves the potential to reduce the variance of estimates (the same strata will be used in every sample, reducing variance in the estimates across hypothetical samples)
- Expect gains in precision over designs that include the between variance as well
- But gains are not guaranteed. The reductions in the variance of estimates depend on the allocation.
- How do we determine how many elements to sample from each stratum?

Allocation

1. Proportionate allocation: Representative sampling where the sample reflects the population with respect to the stratification variable, $n_h/n = N_h/N = W_h$. We have $\pi_{hi} = n_h/N_h = n/N$, i.e., epsem

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4. Optimum allocation: $n_h \propto W_h S_h/\sqrt{J_h}$, where J_h is the cost related to stratum h

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- Design effect:

$$deff = \frac{var(\bar{y}_w)}{var_{SRS}(\bar{y})} = \frac{\frac{1-f}{n} s_w^2}{\frac{1-f}{n} s^2} = \frac{s_w^2}{s^2} = 1 - \frac{\sum_h W_h (\bar{y}_h - \bar{y}_w)^2}{s^2}$$

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- In general deff$<$1

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    - Suppose $S_h = S_{h'}$ for two different strata $h \neq h'$
    - Equal allocation minimized the sampling variance $var(\bar{y}_h - \bar{y}_{h'})$

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  - Large gains require variation among S_h

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- Need estimates of S_h
    - In practice, use reasonable estimates
    - Large gains require variation among S_h
    - Large gains unlikely for proportions

Neyman allocation

- Consider $n_h = kW_hS_h$ with $k = n/\sum_h W_hS_h$
- Smallest sampling variance $var(\bar{y}_w)$
- Gains in precison greater than proportionate allocation
- Need estimates of S_h
    - In practice, use reasonable estimates
    - Large gains require variation among S_h
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    - For multipurpose surveys, allocations will vary as S_h's vary across characteristics

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Allocation: Summary

	Proportionate	Equal	Neyman	Optimum
Goal	Representative	minimize the sampling variance for comparisons $var(\bar{y}_h - \bar{y}_{h'})$	Minimize the sampling variance $var(\bar{y}_w)$	Minimize the sampling variance under total fixed cost
n_h	nW_h	n/H	kW_hS_h	$kW_hS_h/\sqrt{C_h}$
epsem?	Yes	No	No	No
deff	< 1	unsure	< 1	< 1
Multi-purpose	All variables	All variables	Per variable	Per variable

Example R code

```r
### Proportional allocation

popsize <- table(agpop$region)
propalloc <- 300*popsize/sum(popsize); propalloc
```

```
      NC       NE        S        W 
102.7290  21.4425 134.6979  41.1306 
```

```r
# Round to nearest integer
propalloc_int <- round(propalloc); propalloc_int
```

```
 NC  NE   S   W 
103  21 135  41 
```

```r
sum(propalloc_int) # check that stratum sample sizes sum to 300
```

```
[1] 300
```

```r
### Neyman allocation

stratvar <- c(1.1,0.8,1.0,2.0)
# Make sure the stratum variances in stratvar are in same
#  order as the table in popsize
neymanalloc <- 300*(popsize*sqrt(stratvar))/sum(popsize*sqrt(stratvar)); neymanalloc
```

```
       NC        NE         S         W 
101.07640  17.99204 126.36327  54.56828 
```

Example cont.

```
neymanalloc_int <- round(neymanalloc); neymanalloc_int
```

```
 NC  NE   S   W
101  18 126  55
```

```
sum(neymanalloc_int)
```

```
[1] 300
```

```
### Optimal allocation

relcost <- c(1.4,1.0,1.0,1.8)
# Make sure the relative costs in relcost are in same
# order as the table in popsize
optalloc <- 300*(popsize*sqrt(stratvar/relcost))/sum(popsize*sqrt(stratvar/relcost)); optalloc
```

```
       NC        NE         S         W
 94.75776  19.95766 140.16833  45.11626
```

```
optalloc_int <- round(optalloc); optalloc_int
```

```
 NC  NE   S   W
 95  20 140  45
```

```
sum(optalloc_int)
```

```
[1] 300
```

Determining the total sample size

- Define a quantity $v = \sum_h \frac{n}{n_h} (\frac{N_h S_h}{N})^2$ as an "average" variability per unit in a stratified random sample with the specified allocation, similar to S^2 as the variability per unit in an SRS

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- Ignoring all stratum fpcs, $n_0 = z^2_{\alpha/2} v / e^2$ is the required sample size to give the margin of error e
- It can also be calculated as $n_{SRS} v / S^2$, with n_{SRS} being the required SRS sample size
- If $v < S^2$, as in proportional allocation, stratified sampling allows a desired precision with a smaller sample size than SRS

Number of strata

- Stratification requires discrete categories
  - Stratifying variables may be discrete
  - Continuous stratifying variables divided into categories
- How many categories to capture gains possible?
  - Generally 3-6 strata adequate for a single predictor
  - When more than one stratifier, "coarser" cuts on more variables preferred to "finer" cuts
  - "Deepest" stratification for $n_h = 2$ (or $H = n/2$)
  - "Even deeper" stratification: 1 per stratum, a singleton problem

Paired selection

- Paired selections $n_h = 2$ useful in practice
- "Deepest" stratification possible that allows sampling variances to be estimated without assumptions
- Paired selection is epsem: $N_h = N/H$
- Attraction of paired selection is the simplification in variance estimation
- When the design is epsem, proportionately allocated

Paired selection estimation

- The mean is unweighted, and estimate variance under the proportionate allocation: $\bar{y} = \sum_h \sum_i y_{hi}/n = \frac{\sum_h (y_{h1}+y_{h2})}{n}$

$$
\begin{aligned}
var(\bar{y}) &= \frac{1-f}{n} \sum_h W_h s_h^2 (\text{ where } W_h = 2/n) \\
&= \frac{1-f}{n} \sum_h \frac{2}{n} [(y_{h1} - \frac{y_{h1}+y_{h2}}{2})^2 + (y_{h2} - \frac{y_{h1}+y_{h2}}{2})^2] \\
&= \frac{1-f}{n^2} \sum_h (y_{h1} - y_{h2})^2,
\end{aligned}
$$

as the sum of squares of the differences

- For element sampling, the symmetry of the selection and variance estimation disrupted by 1) Blanks in the list, non-responding elements or 2) Analysis of subclasses, Remedy: collapse strata but with overestimated variance

Poststratification

- Poststratification: variables to be used to create strata are not available at the time of selection
- Stratify after selection using variables collected during the survey
- Gains in precision are possible, with suitable modification to variance estimation
- Population control adjustment
- Poststratification requires
  - Poststrata known for each element
  - Poststratum weights W_h for each poststratum
  - New (approximate) variance estimator

Summary

1. Identify stratifying variables correlated with the measure(s) of interest
2. Choose "cuts" on the stratifying variables and divide the population into strata
3. S_h, $deff$, and S^2 are known, at least approximately, and known that fpc's within strata can be ignored
4. Compute an stratified sample size $n = n_{SRS} * deff$
5. Determine an allocation for the desired n
6. Adjust n based on expected $deff$ & allocate
7. Select sample and compute estimates taking the stratified sample selection into account